



PROJECT REPORT

GROUP MEMBERS:

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OCTARECON

Introduction:

OCTARECON is a comprehensive network reconnaissance and information-gathering tool designed for cybersecurity professionals and researchers. The tool integrates multiple functionalities, such as domain and IP analysis, CMS detection, subdomain enumeration, and email harvesting, to aid in detailed reconnaissance tasks.

It is a Python-based command-line utility that combines efficiency and usability. The project leverages modular components for extensibility and performance, ensuring that it can be tailored for various reconnaissance scenarios.

Features:

OCTARECON offers the following key features, each implemented as a modular tool:

1. IP Lookup (I):

- Resolves a domain name to its corresponding IP address.
- Utilizes DNS resolution for accurate results.

2. GeoIP Lookup (G):

- Fetches geographical location details for a given IP address.
- Provides country, city, and ISP details using external APIs.

3. IP Netblock Analysis (N):

- Identifies the network block of the provided IP address.
- Helps determine ownership and the range of IPs within the same block.

4. CMS Detection (C):

- Detects the Content Management System (CMS) used by a website.
- Supports popular CMS platforms such as WordPress, Joomla, and Drupal.

5. Subdomain Enumeration (S):

- Discovers subdomains associated with a domain.
- Useful for identifying additional attack surfaces.

6. Whois Lookup (W):

- Retrieves domain registration details such as registrar, registrant, and expiry date.
- Valuable for assessing domain ownership and validity.

7. SSL Information (L):

- Extracts SSL/TLS certificate details of a domain.
- Assesses security configurations and certificate validity.

8. Email Harvesting (E):

- Identifies email addresses associated with a domain.
- Useful for phishing simulations and contact enumeration.

9. Run All (A):

- Executes all tools in a single operation for comprehensive analysis.

Design Overview

The project is implemented in Python and employs modular programming principles for scalability. Each tool is implemented as a separate module within the tools directory, ensuring code organization and maintainability.

Core Components

1. Tools:

- Each feature is defined in its respective module, such as tools.nslookup, tools.geoip, etc.
- Modules are designed to handle both individual execution and integration within the main program.

2. User Interface:

- Command-line based interface for simplicity and quick interaction.
- ASCII art introduces a visual identity for the tool.

3. Logging:

- Results are saved in a file (finalresult.txt) for later analysis.
- Simultaneous logging to the console and the file is handled using a custom Tee class.

4. Progress Indication:

- A unified progress bar (via the tqdm library) provides real-time feedback on task execution.

5. Error Handling:

- Input validation ensures appropriate tool execution.
- Robust error handling prevents runtime interruptions.

Technical Workflow:

Execution Steps:

1. Display Menu:

- The program prints an ASCII art banner and provides users with a menu of tools to select.

2. Input Handling:

- Users input a domain name and select one or more tools by their corresponding letters.

3. Tool Execution:

- The selected tools are executed sequentially. If "Run All" (A) is chosen, all tools are executed.
- IP resolution is optimized by caching the result for reuse in IP-based tools.

4. Output Logging:

- Results are printed on the console and simultaneously written to a file for persistent storage.

5. Timing Analysis:

- The total execution time is calculated and displayed to the user.

Sample Output:

Terminal Display

```
kali Linux (Snapshot 1) [Running] - Oracle VM VirtualBox
File Actions Edit View Help

(hali@kali): ~/Desktop
└─ $ cd ethical_project
(hali@kali): ~/Desktop/ethical_project
└─ $ python octarecon.py

      OOO  CCCCC  TTTTTT  AAAA  RRRR  EEEE  CCCCC  OOO  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      OOO  CCCCC  T    A  A  R  R  E    EEEE  CCCCC  OOO  N  N

      -----
      This is OCTARECON
      -----

      PRESENTED BY 22K4792, 22K4711 AND 22K4676

Input the domain: nu.edu.pk

Select one or more options to run (e.g., I G W):
I - IP Lookup
G - Geolp
N - IP Netlock
C - OS Detection
5 - SubDomain Enumeration
W - Whois Lookup
L - SSL Info
E - Email Harvesting
A - Run All

Enter your choices (letters): E I C
Overall Progress: 0%
Running Email Harvesting ...

=====
EMAILS FOUND FOR nu.edu.pk : 933
=====

Domain: nu.edu.pk
Email List:
1. Email: hafeez.rehman@nu.edu.pk, Name: Hafeez Ur Rehman, Verification: VALID
2. Email: sadia.nadeem@nu.edu.pk, Name: Sadia Nadeem, Verification: VALID
3. Email: muhammad.naveed@nu.edu.pk, Name: Muhammad Naveed, Verification: UNVERIFIED
4. Email: samia.mahmood@nu.edu.pk, Name: Samia Mahmood, Verification: UNVERIFIED
5. Email: mareez.fazal@nu.edu.pk, Name: Mareez Fazal, Verification: UNVERIFIED
6. Email: umar.niaz@nu.edu.pk, Name: Umar Niaz, Verification: UNVERIFIED
7. Email: omer.saleem@nu.edu.pk, Name: Omer Saleem, Verification: UNVERIFIED

=====
```

```
kali Linux (Snapshot 1) [Running] - Oracle VM VirtualBox
File Actions Edit View Help

(hali@kali): ~/Desktop
└─ $ cd ethical_project
(hali@kali): ~/Desktop/ethical_project
└─ $ python octarecon.py

      OOO  CCCCC  TTTTTT  AAAA  RRRR  EEEE  CCCCC  OOO  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      O  O  C    T    A  A  R  R  E    C  O  O  N  N  N
      OOO  CCCCC  T    A  A  R  R  E    EEEE  CCCCC  OOO  N  N

      -----
      This is OCTARECON
      -----

      PRESENTED BY 22K4792, 22K4711 AND 22K4676

Input the domain: nu.edu.pk

Select one or more options to run (e.g., I G W):
I - IP Lookup
G - Geolp
N - IP Netlock
C - OS Detection
5 - SubDomain Enumeration
W - Whois Lookup
L - SSL Info
E - Email Harvesting
A - Run All

Enter your choices (letters): E I C
Overall Progress: 0%
Running Email Harvesting ...

=====
EMAILS FOUND FOR nu.edu.pk : 933
=====

Domain: nu.edu.pk
Email List:
1. Email: hafeez.rehman@nu.edu.pk, Name: Hafeez Ur Rehman, Verification: VALID
2. Email: sadia.nadeem@nu.edu.pk, Name: Sadia Nadeem, Verification: VALID
3. Email: muhammad.naveed@nu.edu.pk, Name: Muhammad Naveed, Verification: UNVERIFIED
4. Email: samia.mahmood@nu.edu.pk, Name: Samia Mahmood, Verification: UNVERIFIED
5. Email: mareez.fazal@nu.edu.pk, Name: Mareez Fazal, Verification: UNVERIFIED
6. Email: umar.niaz@nu.edu.pk, Name: Umar Niaz, Verification: UNVERIFIED
7. Email: omer.saleem@nu.edu.pk, Name: Omer Saleem, Verification: UNVERIFIED
8. Email: irfan.khalid@nu.edu.pk, Name: Irfan Khalid, Verification: UNVERIFIED
9. Email: aroosa.safdar@nu.edu.pk, Name: Aroosa Safdar, Verification: UNVERIFIED
10. Email: saad.sulaiman@nu.edu.pk, Name: Saad Sulaiman, Verification: UNVERIFIED
11. Email: a.rehman@nu.edu.pk, Name: A Rehman, Verification: UNVERIFIED
12. Email: mareez.fazal@nu.edu.pk, Name: Mareez Fazal, Verification: UNVERIFIED
13. Email: gmo.eric@nu.edu.pk, Name: Gmo Eric, Verification: UNVERIFIED
14. Email: saifullah.tamir@nu.edu.pk, Name: Saifullah Tamir, Verification: UNVERIFIED
15. Email: ra.ozric@nu.edu.pk, Name: Ra Ozric, Verification: UNVERIFIED
16. Email: amara.seemab@nu.edu.pk, Name: Amara Seemab, Verification: UNVERIFIED
17. Email: ehsanulhaq@nu.edu.pk, Name: Ehsanulhaq, Verification: UNVERIFIED
18. Email: waseem.ikram@nu.edu.pk, Name: Waseem Ikram, Verification: UNVERIFIED
19. Email: rana.ayaz@nu.edu.pk, Name: Rana Ayaz, Verification: UNVERIFIED
20. Email: mirza.hameed@nu.edu.pk, Name: Mirza Hameed, Verification: UNVERIFIED
21. Email: wardah.ahmad@nu.edu.pk, Name: Wardah Ahmad, Verification: UNVERIFIED
22. Email: hassan.nasir@nu.edu.pk, Name: Hassan Nasir, Verification: UNVERIFIED
23. Email: rabbia.nasir@nu.edu.pk, Name: Rabbia Nasir, Verification: UNVERIFIED
24. Email: faraz.ahmad@nu.edu.pk, Name: Faraz Ahmad, Verification: UNVERIFIED
25. Email: sana.fatima@nu.edu.pk, Name: Sana Fatima, Verification: UNVERIFIED
26. Email: antif.saif@nu.edu.pk, Name: Antif Saif, Verification: UNVERIFIED
27. Email: tanvir.rehman@nu.edu.pk, Name: Tanvir Rehman, Verification: UNVERIFIED
28. Email: habib.khan@nu.edu.pk, Name: Habib Khan, Verification: UNVERIFIED
29. Email: mussood.abbas@nu.edu.pk, Name: Mussood Abbas, Verification: UNVERIFIED
30. Email: mira.saleem@nu.edu.pk, Name: Mira Saleem, Verification: UNVERIFIED
31. Email: afab.marroof@nu.edu.pk, Name: Afab Marroof, Verification: UNVERIFIED
32. Email: yaseer.nasir@nu.edu.pk, Name: Yaseer Nasir, Verification: UNVERIFIED
33. Email: umar.farooq@nu.edu.pk, Name: Umar Farooq, Verification: UNVERIFIED
34. Email: agha.mureed@nu.edu.pk, Name: Agha Mureed, Verification: UNVERIFIED
35. Email: zeshan.syed@nu.edu.pk, Name: Zeshan Syed, Verification: UNVERIFIED
36. Email: hafa.tariq@nu.edu.pk, Name: Hafa Tariq, Verification: UNVERIFIED
37. Email: munazza.sameed@nu.edu.pk, Name: Munazza Sameed, Verification: UNVERIFIED
38. Email: m.nasir@nu.edu.pk, Name: M Nasir, Verification: UNVERIFIED
39. Email: haider.guliam@nu.edu.pk, Name: Haider Guliam, Verification: UNVERIFIED
40. Email: zeshan.fazal@nu.edu.pk, Name: Zeshan Fazal, Verification: UNVERIFIED
41. Email: mahrukh.nirwan@nu.edu.pk, Name: Mahrukh Nirwan, Verification: UNVERIFIED
42. Email: shah.dost@nu.edu.pk, Name: Shah Dost, Verification: UNVERIFIED
43. Email: irfan.ishaq@nu.edu.pk, Name: Irfan Ishaq, Verification: UNVERIFIED
44. Email: muhammad.abdul@nu.edu.pk, Name: Muhammad Abdul, Verification: UNVERIFIED
45. Email: asir.raheem@nu.edu.pk, Name: Asir Raheem, Verification: UNVERIFIED
46. Email: faisal.hayat@nu.edu.pk, Name: Faisal Hayat, Verification: UNVERIFIED
47. Email: aliene.ahmad@nu.edu.pk, Name: Aliene Ahmad, Verification: UNVERIFIED
48. Email: amna.shabbaz@nu.edu.pk, Name: Amna Shabbaz, Verification: UNVERIFIED
49. Email: ameer.habib@nu.edu.pk, Name: Ameer Habib, Verification: UNVERIFIED
```

```
kali-linux (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali - Desktop/ethical_project

File Actions Edit View Help
980. Email: abdulsahib@nu.edu.pk, Name: None None, Verification: VALID
981. Email: muhammad.adeel@nu.edu.pk, Name: Muhammad Adeel, Verification: VALID
982. Email: sumaira.sarfaraz@nu.edu.pk, Name: Sumaira Sarfaraz, Verification: UNVERIFIED
983. Email: subhan.ullah@nu.edu.pk, Name: Subhan Ullah, Verification: VALID
984. Email: abdulrahman@nu.edu.pk, Name: Abdulrahman Nore, Verification: VALID
985. Email: fahem@nu.edu.pk, Name: Fahem Akhter Rajput, Verification: VALID
986. Email: a.alib@nu.edu.pk, Name: Areez Ali, Verification: VALID
987. Email: muhammad.hussain@nu.edu.pk, Name: Muhammad Hussain, Verification: VALID
988. Email: salman.shoaib@nu.edu.pk, Name: Salman Shoaib, Verification: VALID
989. Email: 1224@nu.edu.pk, Name: None None, Verification: VALID
919. Email: imran.ahmad@nu.edu.pk, Name: Imran Ahmad, Verification: VALID
911. Email: nazhar.abbas@nu.edu.pk, Name: Nazhar Abbas, Verification: VALID
912. Email: zeshan.alib@nu.edu.pk, Name: Zeshan Ali, Verification: VALID
913. Email: muhammad.igbal@nu.edu.pk, Name: Muhammad Iqbal, Verification: UNVERIFIED
914. Email: abdul.saeed@nu.edu.pk, Name: Abdul Saeed, Verification: VALID
915. Email: shoaib.raza@nu.edu.pk, Name: Shoaib Raza, Verification: VALID
916. Email: hina.amjad@nu.edu.pk, Name: Hina Aejad, Verification: VALID
917. Email: muhammad.shan@nu.edu.pk, Name: Muhammad Shan, Verification: VALID
918. Email: muhammad.adnan@nu.edu.pk, Name: Muhammad Adnan, Verification: VALID
919. Email: saba.kamel@nu.edu.pk, Name: Saba Kamel, Verification: VALID
920. Email: shehryar.rashid@nu.edu.pk, Name: Shehryar Rashid, Verification: VALID
921. Email: muhammad.rafi@nu.edu.pk, Name: Muhammad Rafi, Verification: VALID
922. Email: a.alib@nu.edu.pk, Name: A Ali, Verification: VALID
923. Email: muhammad.tahir@nu.edu.pk, Name: Muhammad Tahir, Verification: VALID
924. Email: ammar.javed@nu.edu.pk, Name: Ammar Javed, Verification: UNVERIFIED
925. Email: zaheer.sani@nu.edu.pk, Name: Zaheer Sani, Verification: VALID
926. Email: intiaz.ahmed@nu.edu.pk, Name: Intiaz Ahmad, Verification: VALID
927. Email: ali.haider@nu.edu.pk, Name: Ali Haider, Verification: VALID
928. Email: ifan.alib@nu.edu.pk, Name: Ifan Ali, Verification: UNVERIFIED
929. Email: sara.khan@nu.edu.pk, Name: Sara Khan, Verification: UNVERIFIED
930. Email: fatima.qaisar@nu.edu.pk, Name: Fatima Qaisar, Verification: UNVERIFIED
931. Email: shakel.ahmad@nu.edu.pk, Name: Shakel Ahmad, Verification: UNVERIFIED
932. Email: toba.javed@nu.edu.pk, Name: Toba Javed, Verification: UNVERIFIED
933. Email: haris.kazmi@nu.edu.pk, Name: Haris Kazmi, Verification: UNVERIFIED

Overall Progress: 33%
Running IP Lookup ...

IP FOR DOMAIN : nu.edu.pk
IP: 203.124.44.78
IP Address: 203.124.44.78
Overall Progress: 67%
Running CMS Detection ...

DETECTED TECHNOLOGIES FOR : nu.edu.pk
```

```
kali-linux (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali - Desktop/ethical_project

File Actions Edit View Help

IP FOR DOMAIN : nu.edu.pk
IP: 203.124.44.78
IP Address: 203.124.44.78
Overall Progress: 67%
Running CMS Detection ...

DETECTED TECHNOLOGIES FOR : nu.edu.pk

Technology: Microsoft ASP.NET
Version: 4.8.28219
Categories: Web Framework
More Info: https://whatcms.org/c/Microsoft-ASP-NET

Technology: CloudFlare
Version: N/A
Categories: CDN
More Info: https://whatcms.org/c/CloudFlare

Technology: cdnjs
Version: N/A
Categories: CDN
More Info: https://whatcms.org/c/cdnjs

Technology: jQuery CDN
Version: N/A
Categories: CDN
More Info: https://whatcms.org/c/jquery-CDN

Technology: IIS
Version: 10.0
Categories: Web Server
More Info: https://whatcms.org/c/IIS

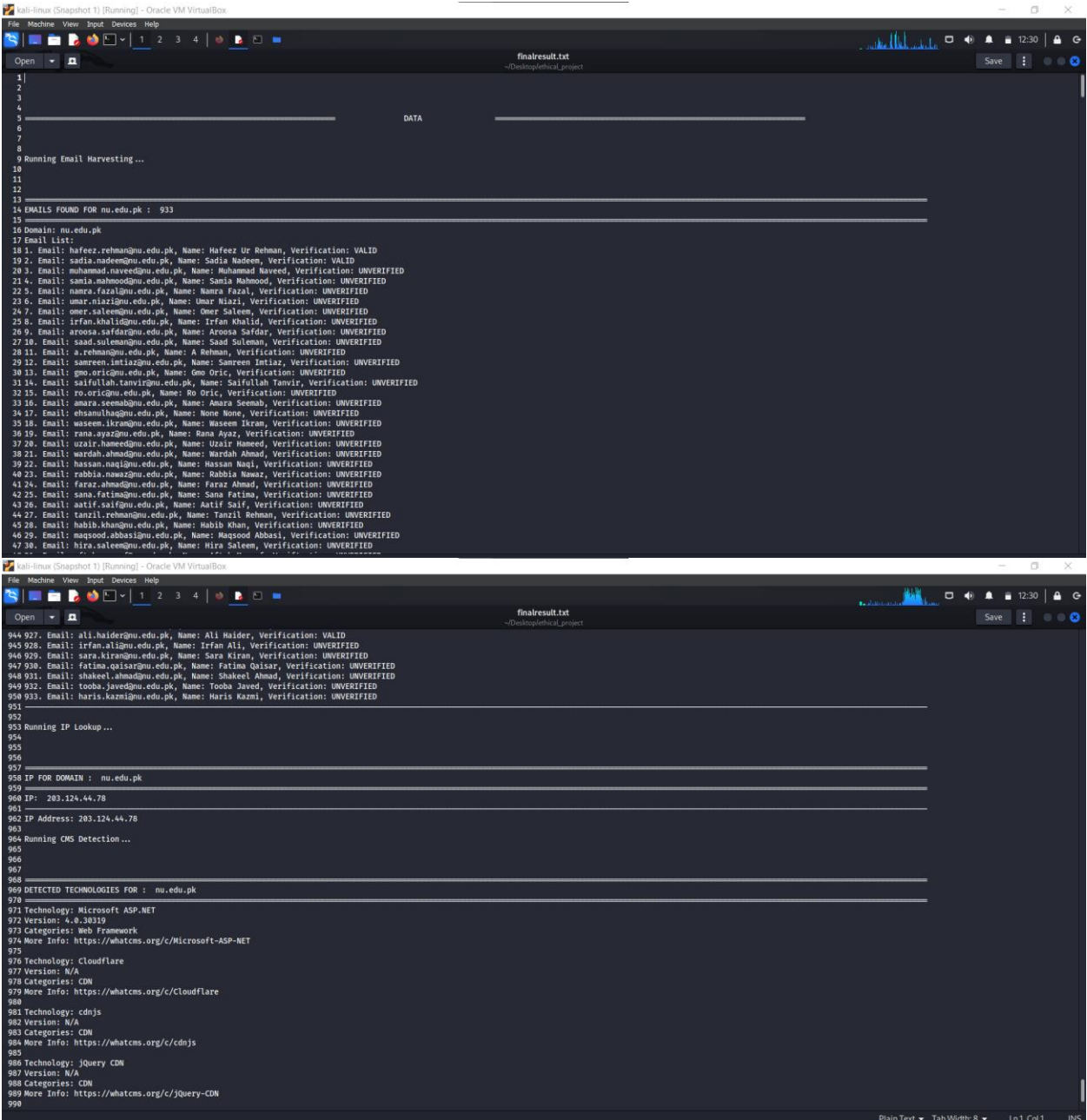
Technology: Windows Server
Version: N/A
Categories: Operating System
More Info: https://whatcms.org/c/Windows-Server

Last Checked: 2024-12-05 07:34:32 UTC

Overall Progress: 100%

TIME TAKEN: 0:00:11.510004
kali@kali: ~/Desktop/ethical_project
```

Saved in finalresult.txt:



Advantages:

- 1. Modularity:** Easy to add or modify tools.
- 2. Automation:** Streamlined execution of multiple tools in a single run.
- 3. Usability:** Clear menu-driven interface with progress tracking.
- 4. Persistence:** Results are saved for later reference.

Future Enhancements:

1. Integration of Additional Tools:

- DNS enumeration and vulnerability scanning could be added.

2. GUI Interface:

- A graphical user interface could enhance usability for non-technical users.

3. Parallel Execution:

- Optimizing performance by executing tools concurrently.

4. Real-time API Support:

- Extend functionality with additional APIs for threat intelligence.

Conclusion:

OCTARECON is a versatile tool that streamlines reconnaissance activities for cybersecurity purposes. Its modular design, combined with robust output logging and user-friendly features, makes it a valuable asset for researchers and professionals. With planned enhancements, it can be extended to address broader security assessment needs.